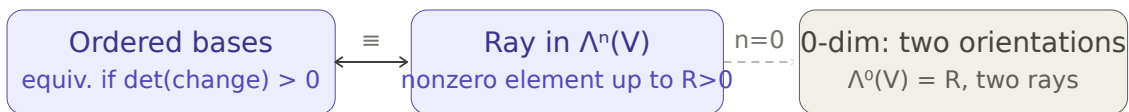
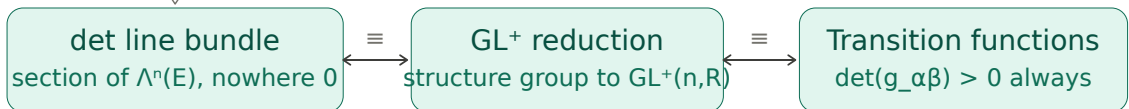


vector spaces



fiberwise

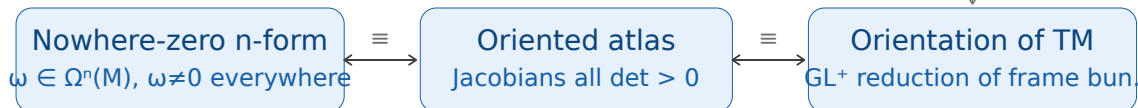
bundles



Short exact sequence: $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$
 $\det(E) \cong \det(E') \otimes \det(E'')$ — orient any two, get the third

TM is a bundle

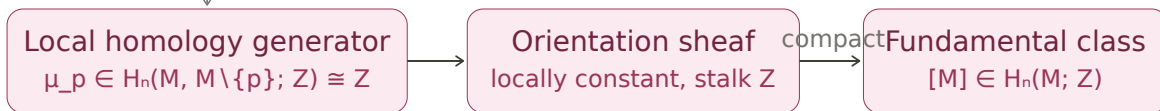
manifolds



Induced orientations (from SES principle)
 ∂M , submanifold normal, product $M \times N$, complex $\rightarrow$ real, preimage

smooth $\rightarrow$ top

global



$[M] \cap - : H^k(M) \rightarrow H_{n-k}(M)$ (Poincaré duality)